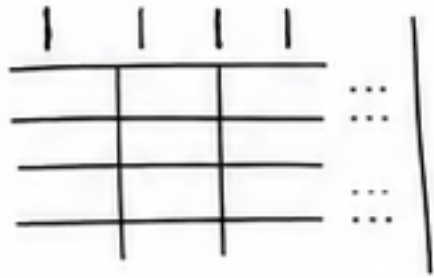


Dimensionality Reduction



df.shape = (150, 5)

↳ 150 rows, 5 columns

1 Feature Selection

Original features (columns)

SL SW PL PW Flower

$x_1 = SL$
 $x_2 = SW$
 $x_3 = PL$
 $x_4 = PW$

From the existing columns, choose only the most useful original columns.

Example:

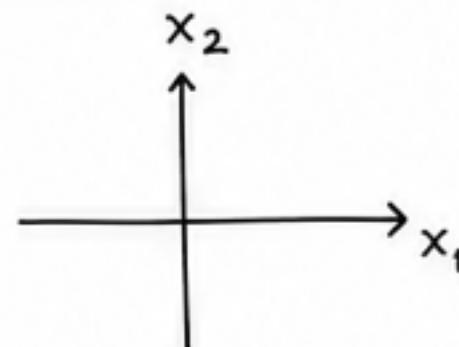
$[SL, SW, PL, PW] \xrightarrow{\text{choose}} [PL, PW]$

✓ Selected only useful original features.

4D
(4 features)



2D
(2 features)



Why dimensionality reduction?

- 1 Visualization
- 2 Compression
- 3 Speed
- 4 Noise removal
- 5 Removing redundancy
- 6 Finding hidden structure

2 Feature Extraction

Create new features by combining the original features.

PC_1, PC_2
↑ ↑
new feature new feature

Feature Selection vs Feature Extraction

Feature Selection

Pick a subset of original features.

$[SL, SW, PL, PW]$



$[PL, PW]$

Keep existing features.

Feature Extraction

Create new features by combining originals.

$[SL, SW, PL, PW]$



PC_1, PC_2

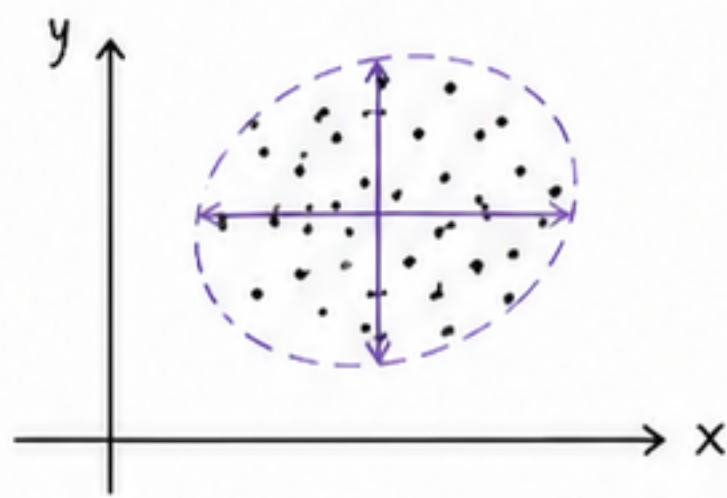
New, transformed features.



Feature selection chooses from existing features.
Feature extraction creates new features.

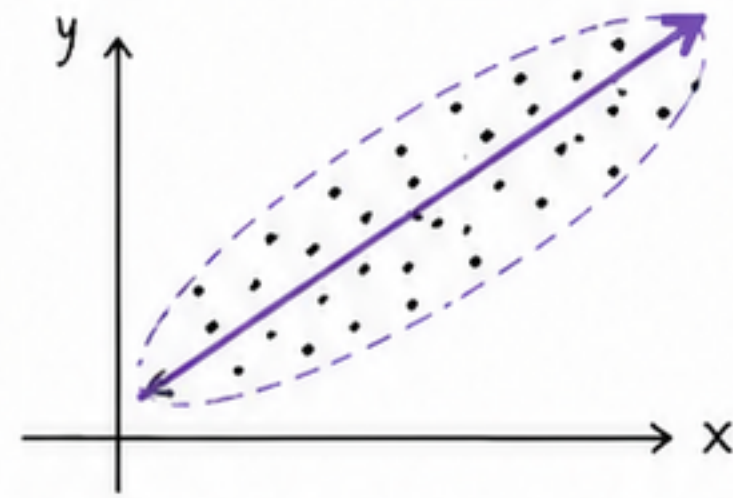
PCA (Principal Component Analysis)

① Eigenvalue (spread of data)



Larger spread $\Rightarrow$ larger eigenvalue.

② Eigenvector (direction of max variance)



Eigenvector points in the direction of maximum variance.

② PCA steps

- ① Center the data
- ② Compute covariance matrix
- ③ Compute eigenvalues and eigenvectors
- ④ Project data onto eigenvectors

Sample data

	H	W	BMI
1	126	78	49
2	128	80	48
3	128	82	50
4	130	82	48
5	130	84	49

Step 1: Mean of each feature

$$\bar{H} = \frac{126 + 128 + 128 + 130 + 130}{5} = 128.4$$

$$\bar{W} = \frac{78 + 80 + 82 + 82 + 84}{5} = 81.2$$

$$\bar{BMI} = \frac{49 + 48 + 50 + 48 + 49}{5} = 48.8$$

Mean vector

$$\mu = \begin{bmatrix} 128.4 \\ 81.2 \\ 48.8 \end{bmatrix}^T$$

Step 2: Center the data

Subtract the mean of each column from the corresponding values.

$$X_c = \begin{bmatrix} H_c & W_c & BMI_c \\ -2.4 & -3.2 & 0.2 \\ -0.4 & -1.2 & -0.8 \\ -0.4 & 0.8 & 1.2 \\ 1.6 & 0.8 & -0.8 \\ 1.6 & 2.8 & 0.2 \end{bmatrix}$$

Check:
Column means of X_c are approximately 0.



Step 3: Covariance matrix

With $n = 5$ samples and X_c centered data.

$$S = \frac{1}{n-1} X_c^T X_c \rightarrow \text{Unbiased covariance (divide by } n-1)$$

Covariance matrix

$$S = \begin{bmatrix} 2.8 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.3 \\ -0.4 & 0.3 & 0.7 \end{bmatrix}$$

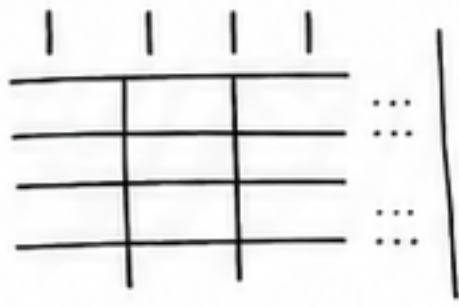
Properties:

- Symmetric: $S = S^T$
- Positive semi-definite



Next step: compute eigenvalues and eigenvectors of S and project data onto eigenvectors (principal components).

Finding Eigenvalues



Eigenvalues of the covariance matrix tell us how much variance each principal component explains.

1 Covariance Matrix

$$S = \begin{pmatrix} 2.8 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.3 \\ -0.4 & 0.3 & 0.7 \end{pmatrix}$$

Identity Matrix

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(3 × 3 identity matrix)

2 Characteristic Equation

$$|S - \lambda I| = 0$$

We subtract λ along the diagonal ($S - \lambda I$) and set the determinant to zero.

$$S - \lambda I = \begin{pmatrix} 2.8 - \lambda & 3.4 & -0.4 \\ 3.4 & 5.2 - \lambda & 0.3 \\ -0.4 & 0.3 & 0.7 - \lambda \end{pmatrix}$$

$$\Rightarrow |S - \lambda I| = 0$$

3 Solving the Determinant Equation

Expanding the determinant and simplifying gives the characteristic polynomial:

$$-\lambda^3 + 8.70\lambda^2 - 8.57\lambda + 0.19 = 0$$

Solving this cubic equation yields the eigenvalues (approximately):

$$\lambda_1 \approx 7.61 \rightarrow \text{Largest eigenvalue}$$

$$\lambda_2 \approx 1.07 \rightarrow \text{Second largest}$$

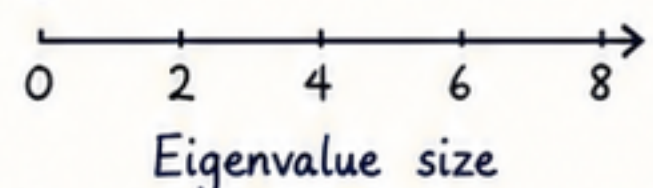
$$\lambda_3 \approx 0.02 \rightarrow \text{Smallest}$$

Eigenvalues (ranked)

- $\lambda_1 \approx 7.61$

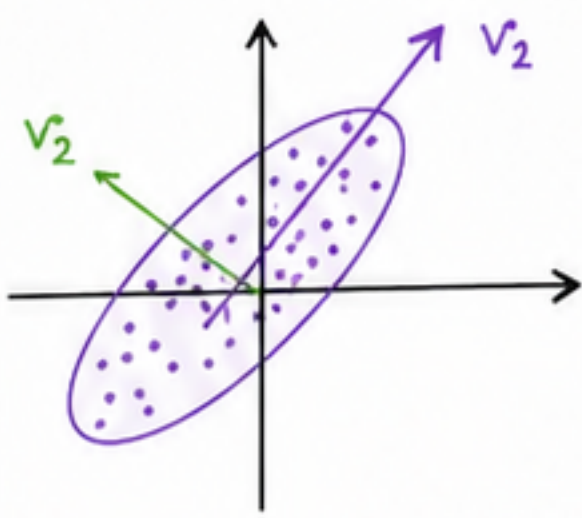
- $\lambda_2 \approx 1.07$

- $\lambda_3 \approx 0.02$



Larger eigenvalue $\Rightarrow$ more variance explained.

Finding Eigenvectors



Key idea:

For each eigenvalue λ , solve

$$(S - \lambda I) v = 0$$

S : covariance matrix

λ : eigenvalue

v : eigenvector

1 Covariance Matrix

$$S = \begin{bmatrix} 2.6 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.1 \\ -0.4 & 0.1 & 0.7 \end{bmatrix} \leftarrow (\text{symmetric})$$

2 Example: Solve for $\lambda_1 \approx 6.00$

$$(S - \lambda_1 I) v = 0$$

$$\left(\begin{bmatrix} 2.6 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.1 \\ -0.4 & 0.1 & 0.7 \end{bmatrix} - \begin{bmatrix} 6.00 & 0 & 0 \\ 0 & 6.00 & 0 \\ 0 & 0 & 6.00 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\left(\begin{bmatrix} -3.40 & 3.40 & -0.40 \\ 3.40 & -0.80 & 0.10 \\ -0.40 & 0.10 & -5.30 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Solving gives a direction proportional to $(-1.1, -1.6, -0.002)^T$

Normalization

Eigenvectors are scaled to unit length:

$$\|v\| = 1$$

The sign of an eigenvector can flip and still represent the same principal direction.

3 Eigenvectors (Normalized)

The normalized eigenvectors (principal directions) are:

$$v_1 \text{ (largest } \lambda) \rightarrow v_1 \approx \begin{bmatrix} -0.577 \\ -0.816 \\ -0.002 \end{bmatrix}$$

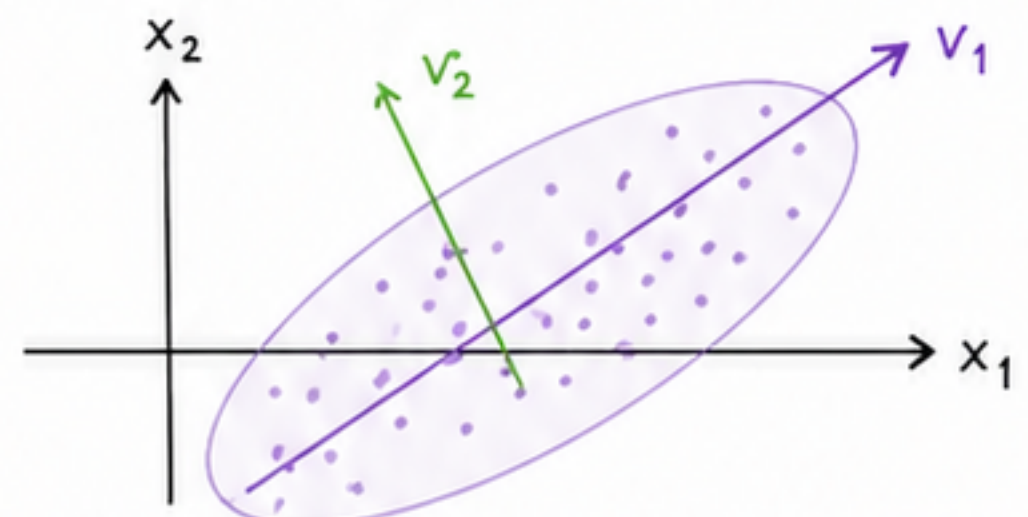
$$v_2 \text{ (second largest } \lambda) \rightarrow v_2 \approx \begin{bmatrix} -0.487 \\ 0.342 \\ 0.804 \end{bmatrix}$$

$$v_3 \text{ (smallest } \lambda) \rightarrow v_3 \approx \begin{bmatrix} -0.656 \\ 0.465 \\ -0.595 \end{bmatrix}$$

★ These vectors form an orthonormal basis:
 $V^T V = I$, $\|v_i\| = 1$, $v_i^T v_j = 0$ ($i \neq j$)

4 Geometric View

Eigenvectors show the principal directions of the data cloud.



- v_1 : direction of maximum variance
- v_2 : second principal direction
- v_3 : third principal direction (out of plane)



Use these eigenvectors to project data onto principal components:

$$Z = X_c V$$

1	1	1	...
			...
			...
			...

Explained Variance and Principal Components

`df.shape = (150, 5)`

↳ 150 rows, 5 columns

1 Eigenvalues (Variance Captured)

Eigenvalues show how much variance each principal component captures.

Component	Eigenvalue (λ)
PC ₁	$\lambda_1 \approx 7.61$
PC ₂	$\lambda_2 \approx 1.07$
PC ₃	$\lambda_3 \approx 0.02$

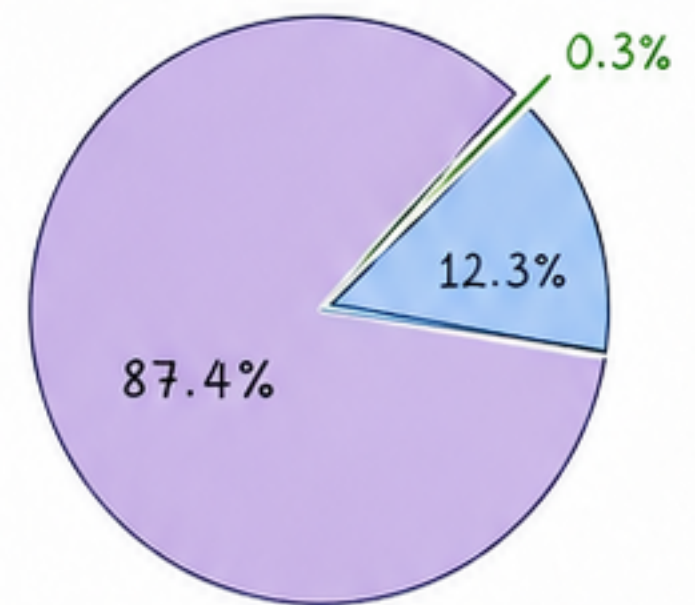
Total variance ≈ 8.70

2 Explained Variance

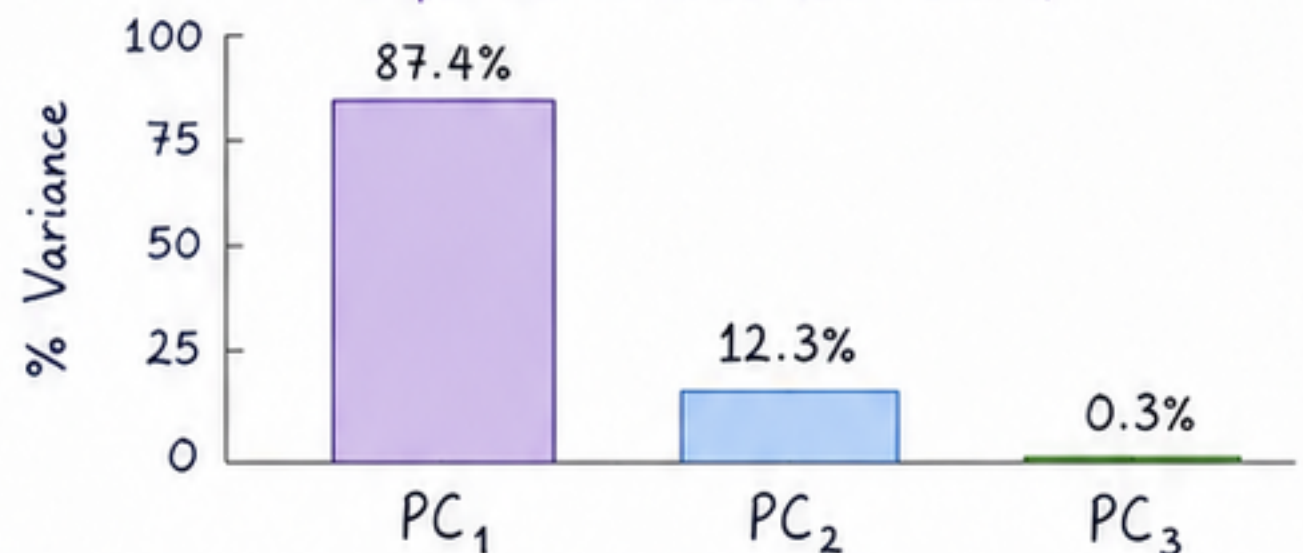
The proportion of variance explained by each principal component.

Component	Explained Variance
PC ₁	$\approx 87.4\%$
PC ₂	$\approx 12.3\%$
PC ₃	$\approx 0.3\%$

Explained Variance



Explained Variance (Bar Chart)



3 Explained Variance Calculation

Explained variance (%) for each PC is:

$$\frac{\lambda_i}{\text{Total Variance}} \times 100$$

- PC₁ : $\frac{7.61}{8.70} \times 100 \approx 87.4\%$
- PC₂ : $\frac{1.07}{8.70} \times 100 \approx 12.3\%$
- PC₃ : $\frac{0.02}{8.70} \times 100 \approx 0.3\%$



PC1 explains most of the variance, so keeping PC₁ and PC₂ is usually enough here.

4 Principal Components (Directions)

Eigenvectors give the directions of the principal components.

We form the transformation matrix using the first two principal components.

Transformation Matrix (first 2 PCs)

$$W = [v_1 \ v_2] =$$

$$\begin{bmatrix} -0.577 & -0.487 \\ -0.816 & 0.342 \\ -0.002 & 0.804 \end{bmatrix}$$

Note



PC1 and PC₂ are new features created from the original H, W, BMI features.



Key takeaway: PC1 captures the most information. Using the top 2 PCs reduces dimensionality while keeping most of the important variance.

Projecting the Data and the Main Idea of PCA

Main steps of PCA

- 1 Center the data
- 2 Compute covariance matrix
- 3 Compute eigenvalues and eigenvectors
- 4 Form PC_1 , PC_2 , PC_3 and project the data

Key formulas

$$X_c = X - \bar{X} \quad \leftarrow \text{center the data (subtract the mean)}$$

$$S = \frac{1}{n-1} X_c^T X_c \quad \leftarrow \text{covariance matrix (n features)}$$

$$Z = X_c W \quad \leftarrow \text{project centered data onto principal components}$$



PCA is an unsupervised algorithm. It does not use the target column. It finds the directions in the feature space where the data has maximum variance.

Projected data (2D: PC_1 , PC_2)

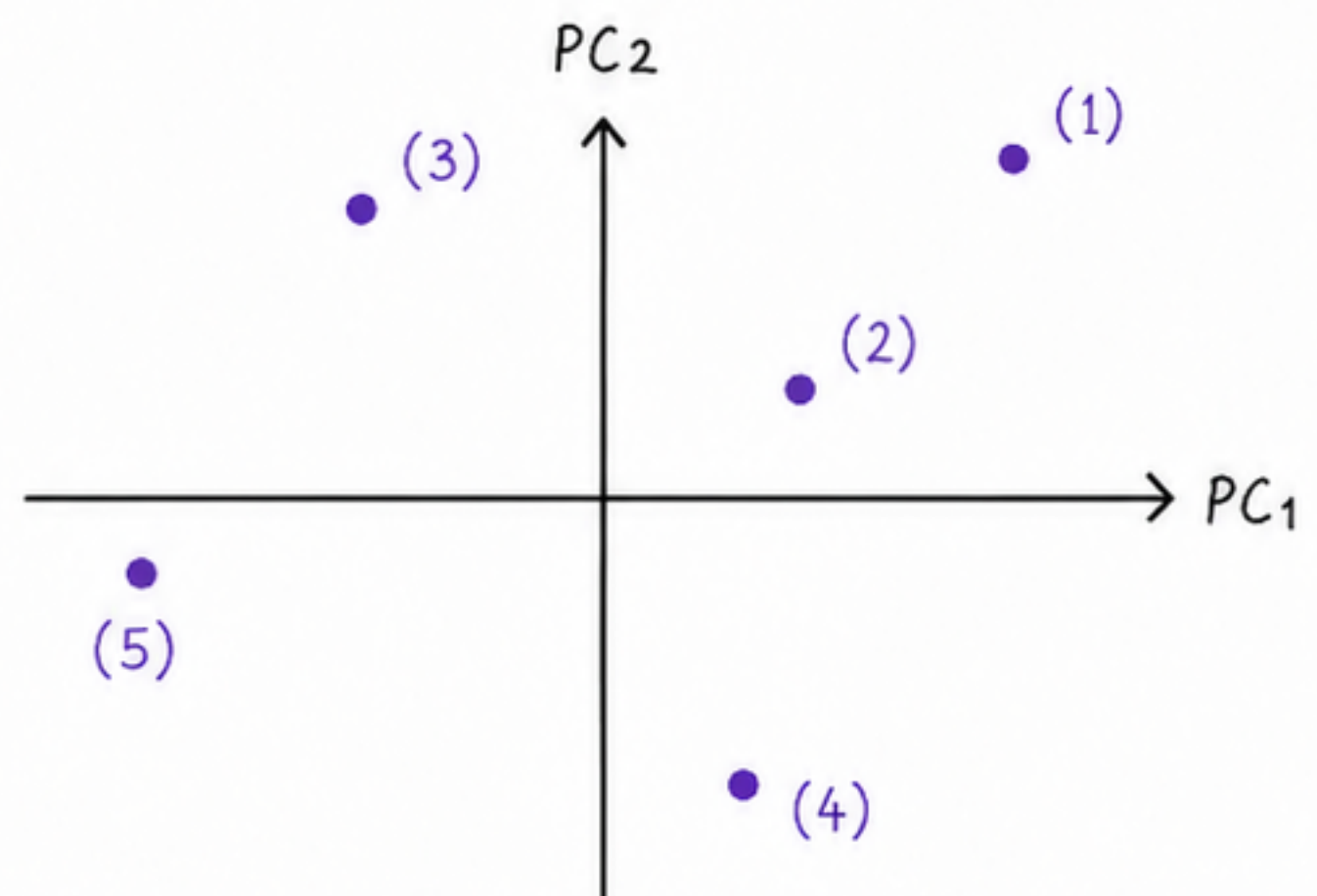
After projecting the centered data onto the top 2 principal components:

Sample	PC1	PC2
1	3.998	0.234
2	1.212	-0.859
3	-0.425	1.433
4	-1.575	-1.148
5	-3.210	0.340

(Values shown are approximate.)

Projection on the PC_1 - PC_2 plane

Each point below is the 2D representation of a sample in the new space.



PCA finds new axes (principal components) where the data spreads the most. Projecting the data onto these axes keeps the most important information in fewer dimensions.

1	1	1	1	...
				...
				...
				...

Dimensionality Reduction Methods



Dimensionality reduction is useful for classification, clustering, and visualization.

Method	Type	Idea	Best use	Visual / Sketch
1 PCA	Linear	Finds new axes with maximum variance.	Feature reduction, preprocessing, visualization.	
2 Isomap	Non-linear	Geometric mapping that preserves geodesic distances.	Unfolding manifold data such as swiss roll.	
3 LLE (Locally Linear Embedding)	Non-linear	Preserves local neighborhood structure.	Manifold learning / local structure.	
4 t-SNE (t-distributed stochastic neighbor embedding)	Non-linear	Keeps similar points close in a low-dimensional map.	Mainly 2D or 3D visualization of clusters.	



Linear methods (e.g., PCA) are simple and fast.
Non-linear methods capture complex manifold structures.